Numerical Analysis Quiz Name: _____

Question 1. Let f be sufficiently smooth and $h > 0$. Consider

$$D_h f(x) = \frac{af(x) + bf(x+h) + cf(x+3h)}{h^2}.$$

Determine the coefficients a, b, and c so that $D_h f(x)$ is a consistent approximation to $f''(x)$. Justify your answer using Taylor expansions about x.

Solution.

We seek coefficients a, b, and c such that

$$D_h f(x) = \frac{af(x) + bf(x+h) + cf(x+3h)}{h^2}$$

is a consistent approximation to $f''(x)$.

Using Taylor expansions about x, we have

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + O(h^4),$$

and

$$f(x+3h) = f(x) + 3hf'(x) + \frac{9h^2}{2}f''(x) + \frac{27h^3}{6}f'''(x) + O(h^4).$$

Substituting into the expression for $D_h f(x)$ gives

$$\begin{aligned} D_h f(x) = \frac{a+b+c}{h^2}f(x) + \frac{b+3c}{h}f'(x) \\ + \frac{b+9c}{2}f''(x) + \frac{b+27c}{6}hf'''(x) + O(h^2). \end{aligned}$$

For consistency with $f''(x)$, we require

$$\begin{cases} a+b+c = 0, \\ b+3c = 0, \\ b+9c = 2. \end{cases}$$

Subtracting the second equation from the third gives

$$6c = 2, \qquad \text{so} \qquad c = \frac{1}{3}.$$

Then

$$b = -3c = -1, \qquad a = -b - c = \frac{2}{3}.$$

Therefore, the finite difference formula is

$$\boxed{D_h f(x) = \frac{2f(x) - 3f(x+h) + f(x+3h)}{3h^2}.}$$

Moreover, the leading truncation error is

$$\begin{aligned} D_h f(x) - f''(x) &= \frac{b+27c}{6}hf'''(x) + O(h^2) \\ &= \frac{4}{3}hf'''(x) + O(h^2). \end{aligned}$$

Thus,

$$\boxed{D_h f(x) - f''(x) = \frac{4}{3} h f'''(x) + O(h^2).}$$

The approximation is first-order accurate. □

Question 2. Let f be a sufficiently smooth function and let $h > 0$. Consider the following approximation to $f'(x)$:

$$D_h f(x) = \frac{-9f(x-h) + 8f(x) + f(x+3h)}{12h}.$$

Using Taylor expansions with Lagrange remainders, show that there exists $c \in (x-h, x+3h)$ such that

$$D_h f(x) - f'(x) = \frac{1}{2} h^2 f'''(c).$$

Solution.

We consider

$$D_h f(x) = \frac{-9f(x-h) + 8f(x) + f(x+3h)}{12h}.$$

Using Taylor expansions about x with Lagrange remainders, we have

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2} f''(x) - \frac{h^3}{6} f'''(c_1),$$

where $c_1 \in (x-h, x)$, and

$$f(x+3h) = f(x) + 3hf'(x) + \frac{9h^2}{2} f''(x) + \frac{27h^3}{6} f'''(c_2),$$

where $c_2 \in (x, x+3h)$.

Substituting into the expression for $D_h f(x)$, we obtain

$$\begin{aligned} D_h f(x) &= f'(x) + \frac{h^2}{12} \left(\frac{9}{6} f'''(c_1) + \frac{27}{6} f'''(c_2) \right) \\ &= f'(x) + \frac{h^2}{8} f'''(c_1) + \frac{3h^2}{8} f'''(c_2). \end{aligned}$$

Therefore,

$$D_h f(x) - f'(x) = \frac{h^2}{2} \left(\frac{1}{4} f'''(c_1) + \frac{3}{4} f'''(c_2) \right).$$

Since f''' is continuous on $[x-h, x+3h]$, the intermediate value theorem implies that there exists $c \in (x-h, x+3h)$ such that

$$f'''(c) = \frac{1}{4} f'''(c_1) + \frac{3}{4} f'''(c_2).$$

Consequently,

$$\boxed{D_h f(x) - f'(x) = \frac{1}{2} h^2 f'''(c), \qquad c \in (x-h, x+3h).}$$

Thus, the approximation is second-order accurate. □